CE 400/500
Special Topics: AI for Chemical Engineers

Lecture 1: Introduction

University at Buffalo
The State University of New York

Organization

- **Instructor:**

  Kaihang Shi

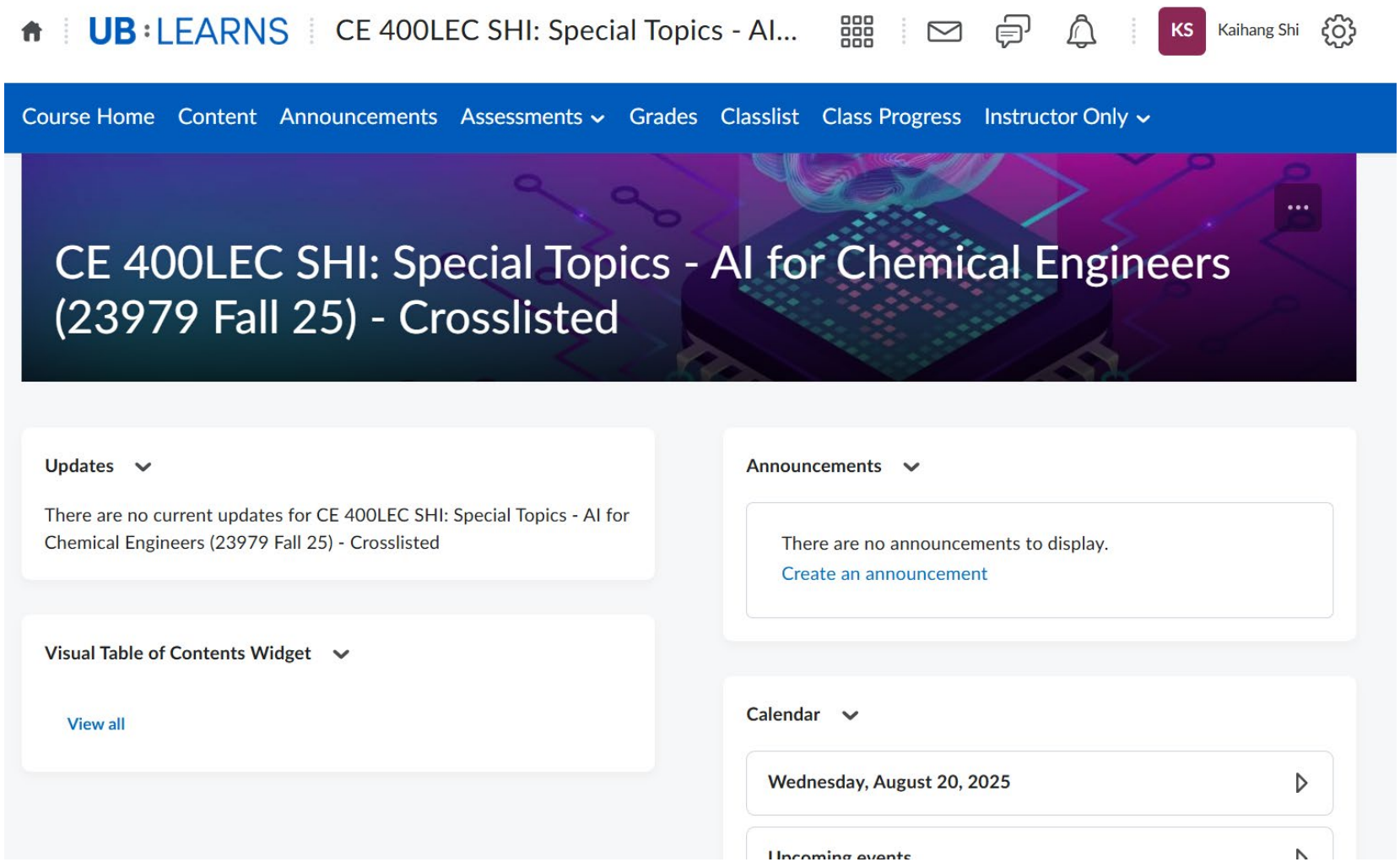
  509 Furnas Hall

  Phone: 716-645-2615

  Email: kaihangs@buffalo.edu

  Office Hours: by email appointment in my office or via Zoom

UB Brightspace

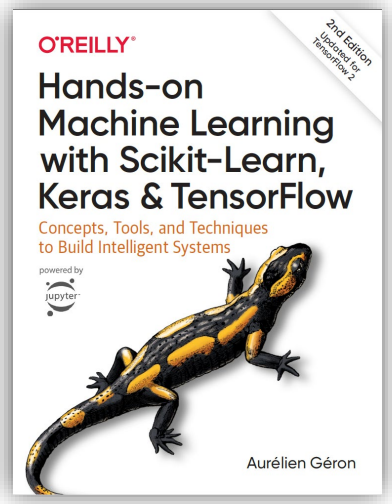


Course Objectives

- Use Python and large language models (LLMs) to efficiently write, debug, and solve simple coding problems relevant to chemical engineering.
- Understand core AI techniques—Predictive (regression, random forests, neural networks), Prescriptive (genetic algorithms, Bayesian optimization), and Generative models (LLMs)—and apply them to chemical engineering problems.
- Translate chemical engineering challenges into clear AI problem statements, facilitating effective collaboration with AI practitioners.
- Critically evaluate basic AI methods and outputs, recognizing their strengths, limitations, and ethical implications, ensuring alignment with engineering standards.
- Leverage generative AI tools (mainly LLM) to enhance productivity and innovation in chemical engineering.

Recommended Textbook

- This course does not require a traditional textbook due to the rapidly evolving nature of artificial intelligence and machine learning.
- Instead, we will rely on current research papers, online resources, and real-time case studies to ensure you are learning the most up-to-date concepts and practices in the field.
- All recommended readings and materials will be provided through the lecture slides and the course management system such as UB Learns.
- Students interested in deeper technical implementation may find “Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow” by Aurélien Géron to be a valuable reference. This book provides practical, beginner-friendly examples that complement ML techniques that will be discussed in this course. **eBook access is available via UB library.**



Grading

- **Final project presentation (20%) and written report (15%)**
  - Every student is required to complete a course project. Topic is flexible – reformulate a (ChemE-related) problem you are interested in as an AI/ML problem.
  - Detailed guidelines will be provided in UB Learns.
  - Oral presentation will be at the end of semester. Written report is due one week after.

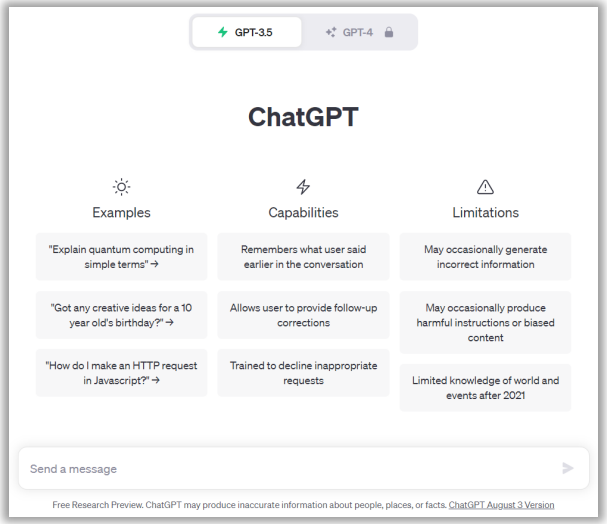
- **Mid-term presentation (25%)**
  - Journal article review on applications of AI/ML in chemical & biological engineering
  - 15 minutes + 5 minutes Q&A (see syllabus for details)
  - Tentative date is 10/9 (Thursday)

- **Homework assignments (20%)**
- **Attendance and in-class discussion (20%)**

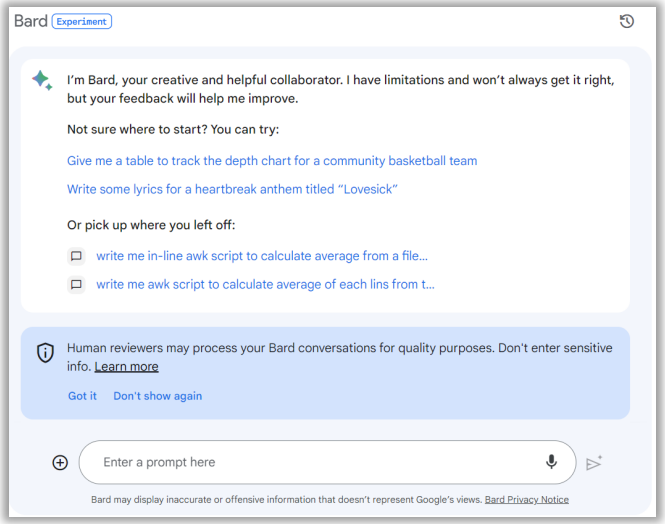
The grade will reflect different expectations for undergrads and graduate students.

Use of Generative AI services

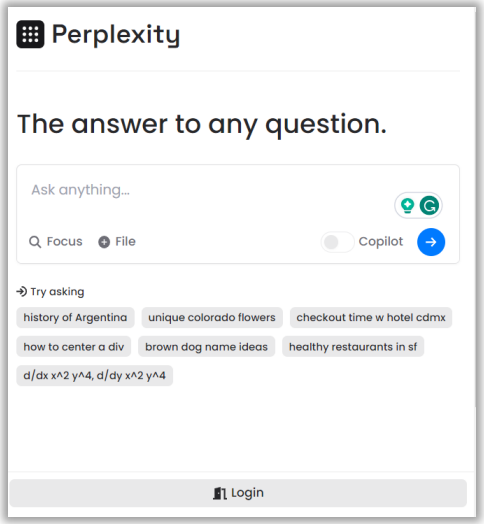
ChatGPT



Google Gemini

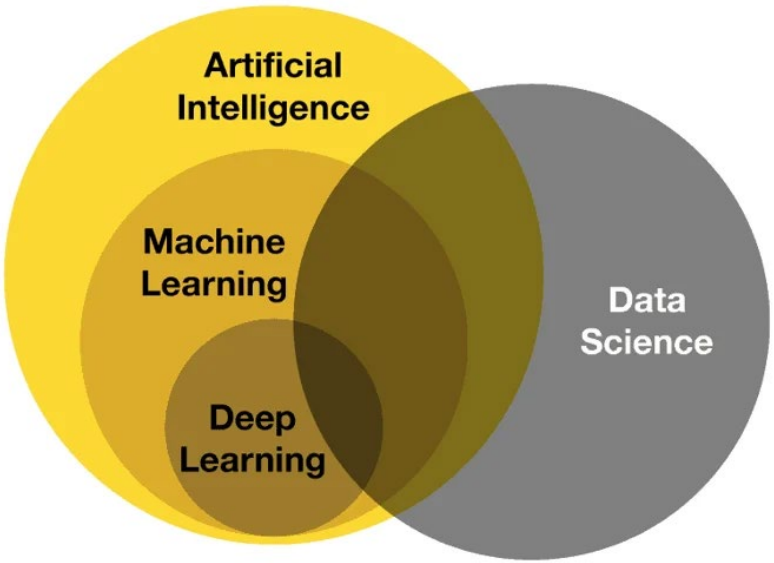


Perplexity



- **You are encouraged** to use AI/ML technology in this class to help you complete homework assignments, projects, and to enhance your learning experience.
- However, directly copying solutions word-for-word from these services is **strictly prohibited** and will be considered as **Falsifying Academic Materials**.

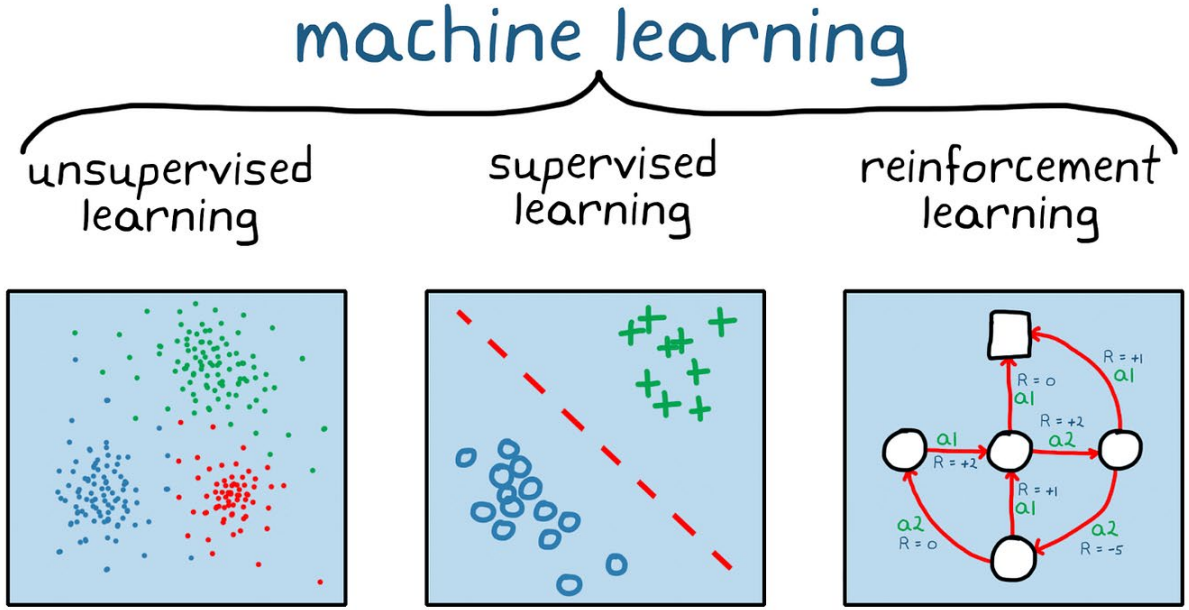
Relationship of AI, ML, and Data Science



- **Artificial Intelligence (AI):** The broad concept of developing machines that can simulate human thinking, reasoning, and behavior. (Siri, chatbots etc.)
- **Machine Learning (ML):** A subset of AI where computer systems learn from data to improve their performance (e.g., Netflix recommendation algorithm). *All ML is AI, but not all AI is ML.*
- **Deep Learning (DL):** A subset of ML using artificial neural networks with multiple layers to extract complex patterns from raw data. (e.g., image recognition)
- **Data Science:** The process of extracting insights and patterns from data using statistical analysis and ML techniques to predict future events. (e.g., sports analytics)

https://www.deviq.io/insights/artificial-intelligence-vs-machine-learning-vs-data-science

How ML Learns?

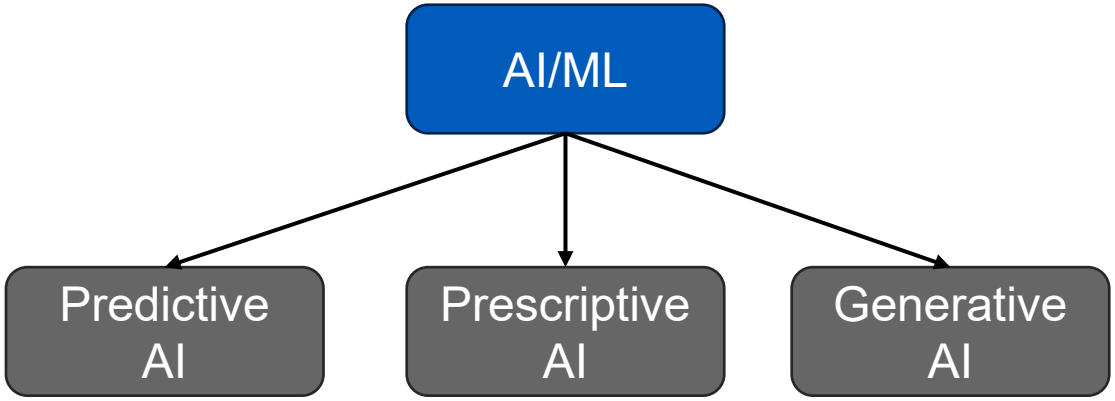


Computer scientists usually classify AI/ML by how models learn from data:

- Supervised Learning: Learning from labelled data (both x and y).
- Unsupervised Learning: Discovering patterns in unlabeled data (only x).
- Reinforcement Learning: Learning through interactions with an environment.

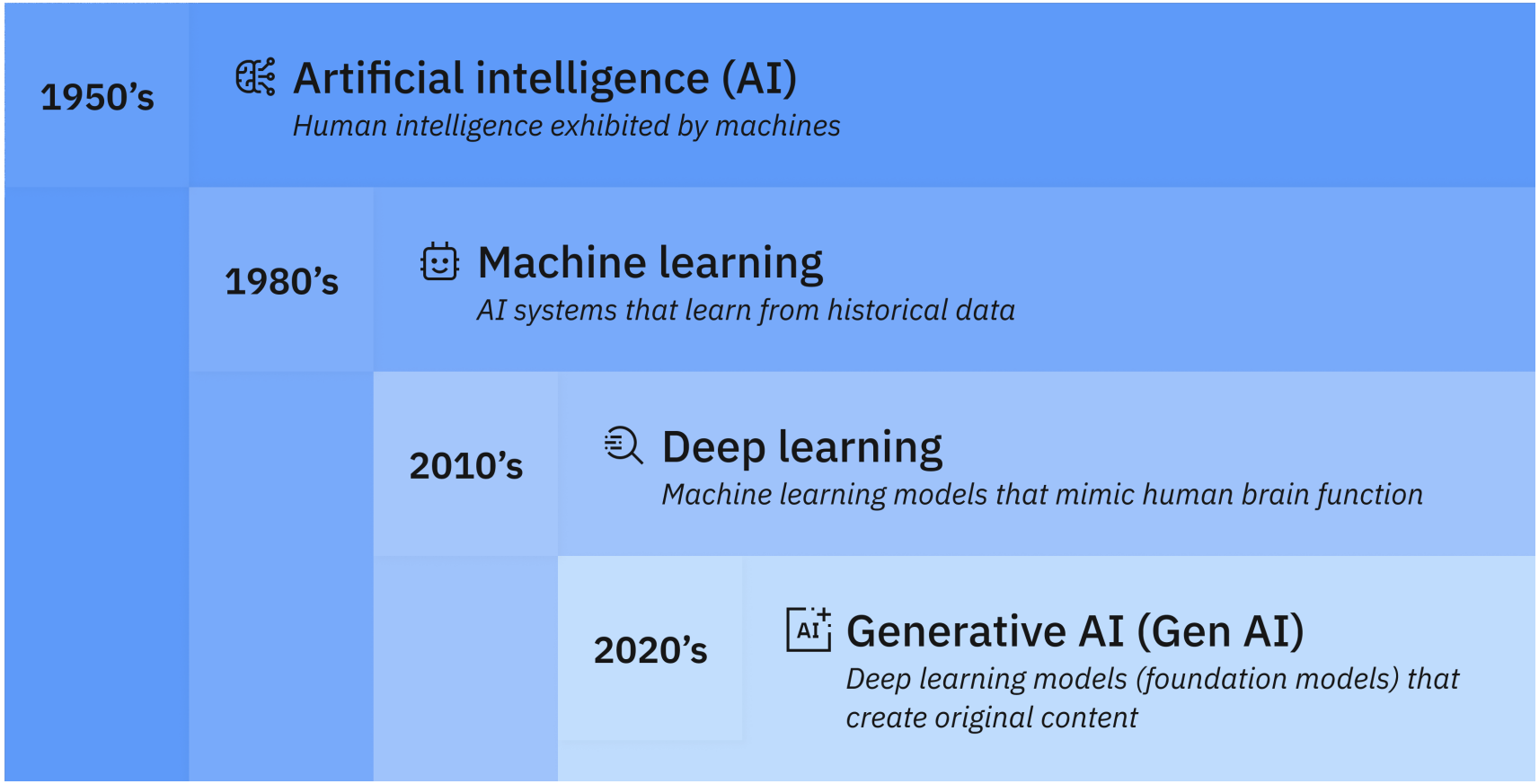
https://medium.com/@anabelenmanjavacas/supervised-unsupervised-and-reinforcement-learning-d0a19d016573
https://www.geeksforgeeks.org/machine-learning/supervised-vs-reinforcement-vs-unsupervised/#

Types of AI/ML to be Covered in This Course: What Can AI Do for Us?



- **Predictive AI:** Forecasts what will happen based on historical data.
  - E.g., weather forecasting systems, Netflix recommendations, prediction of reactor yield, prediction of thermodynamic property
- **Prescriptive AI:** Recommends the best action to take to achieve a goal.
  - E.g., reactor optimization, material performance optimization.
- **Generative AI:** Creates new content that didn't exist before.
  - E.g., ChatGPT, image generation, generating process flow diagrams, generating new molecular structures for drug design, generate safety protocols

Overview of the History of AI



A simple way to think about AI is as a series of nested or derivative concepts that have emerged over more than 70 years

https://www.ibm.com/think/topics/history-of-artificial-intelligence

1950: The Inception

- Alan Turing published Computing Machinery and Intelligence. In this paper, Turing, famous for breaking the German ENIGMA code during WWII and often referred to as the “father of computer science”, asked the following question: “Can machines think?”

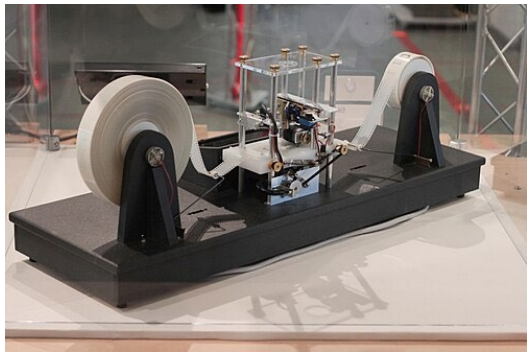
- From there, he offered a test, now famously known as the “Turing Test,” where a human interrogator would try to distinguish between a computer and human text response. While this test has undergone much scrutiny since it was published, it remains an important part of the history of AI, and an ongoing concept within philosophy as it uses ideas around linguistics.

Turing in 1951
(1912 – 1954, aged 41)

https://www.ibm.com/think/topics/artificial-intelligence
https://en.wikipedia.org/wiki/Alan_Turing

More about Alan Turing

- Pioneered the concept of the modern computer and developed the Turing machine, a foundational model for computation.
- Played a crucial role in breaking the German Enigma code during World War II, significantly aiding the Allied victory.
- Developed the concept of “Artificial Intelligence” and proposed the Turing Test to measure a machine’s ability to exhibit intelligent behavior.
- Contributed to mathematics, logic, cryptanalysis, and biology.
- Faced persecution due to his homosexuality and died in 1954.



A physical Turing machine model

Turing in 1930s
(1912 – 1954, aged 41)

https://en.wikipedia.org/wiki/Turing_machine
https://en.wikipedia.org/wiki/Alan_Turing

1956: AI Term Coined

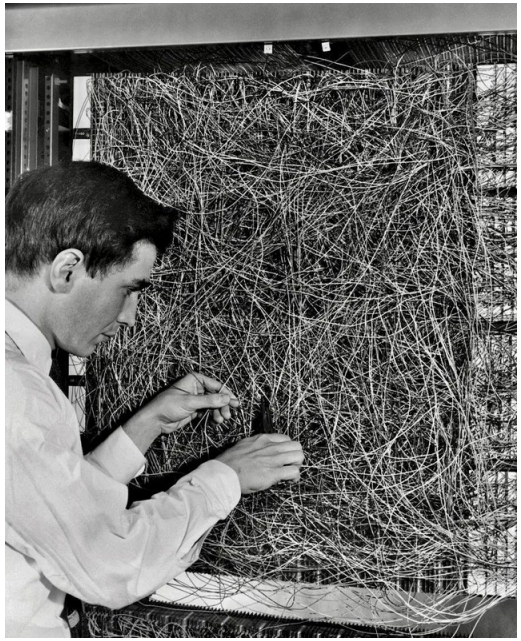
In the back row from left to right are Oliver Selfridge, Nathaniel Rochester, Marvin Minsky, and John McCarthy. In front on the left is Ray Solomonoff; on the right, Claude Shannon. The identity of the person between Solomonoff and Shannon remained a mystery for some time. THE MINSKY FAMILY

- John McCarthy coined the term “artificial intelligence” at the first-ever AI conference at Dartmouth College. (McCarthy went on to invent the Lisp language.)
- Later that year, Allen Newell, J.C. Shaw and Herbert Simon created the *Logic Theorist*, the first-ever running AI computer program.

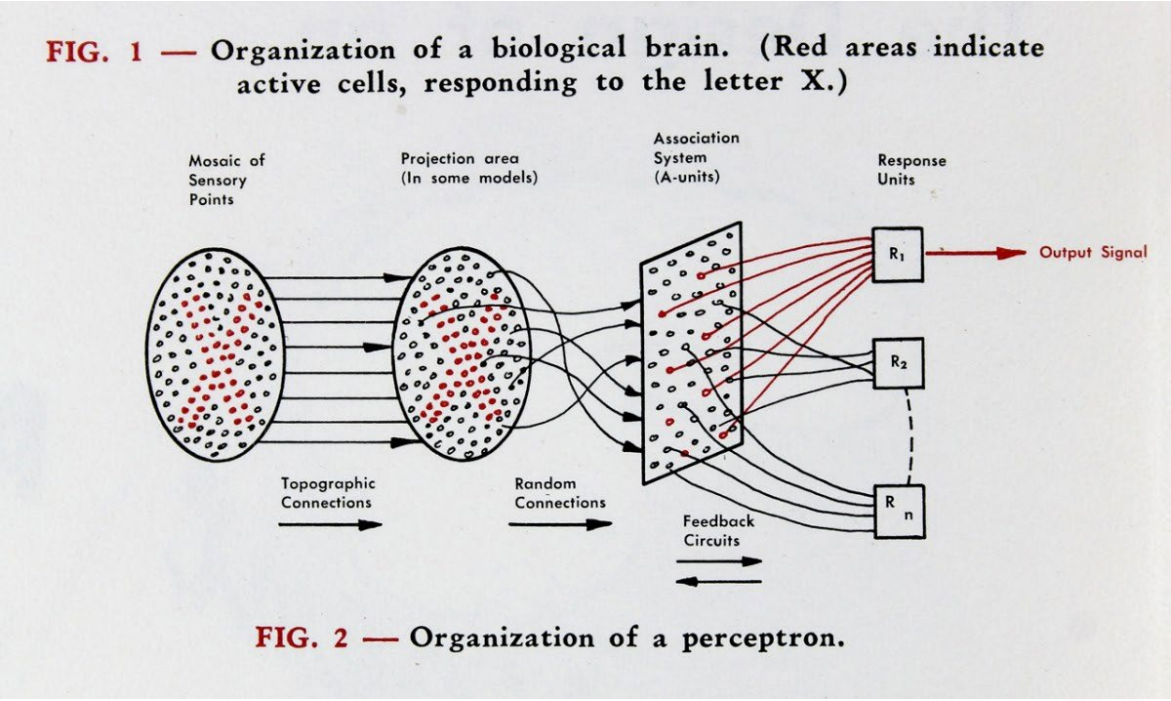
https://www.ibm.com/think/topics/artificial-intelligence

https://spectrum.ieee.org/dartmouth-ai-workshop

1957: The Inception of Neural Network



Frank Rosenblatt with a Mark I Perceptron computer in 1960



An image of the perceptron from Rosenblatt's “The Design of an Intelligent Automaton,” Summer 1958.

- Frank Rosenblatt built the Mark 1 Perceptron, the first computer based on a neural network that “learned” through trial and error.
- The work was first physical implementation of neural network.
- The Mark I Perceptron machine was built at the Cornell Aeronautical Laboratory (now Calspan Corporation) in **Buffalo**, New York.

https://www.ibm.com/think/topics/history-of-artificial-intelligence

https://news.cornell.edu/stories/2019/09/professors-perceptron-paved-way-ai-60-years-too-soon

1986: Backpropagation for training neural networks

Learning representations by back-propagating errors

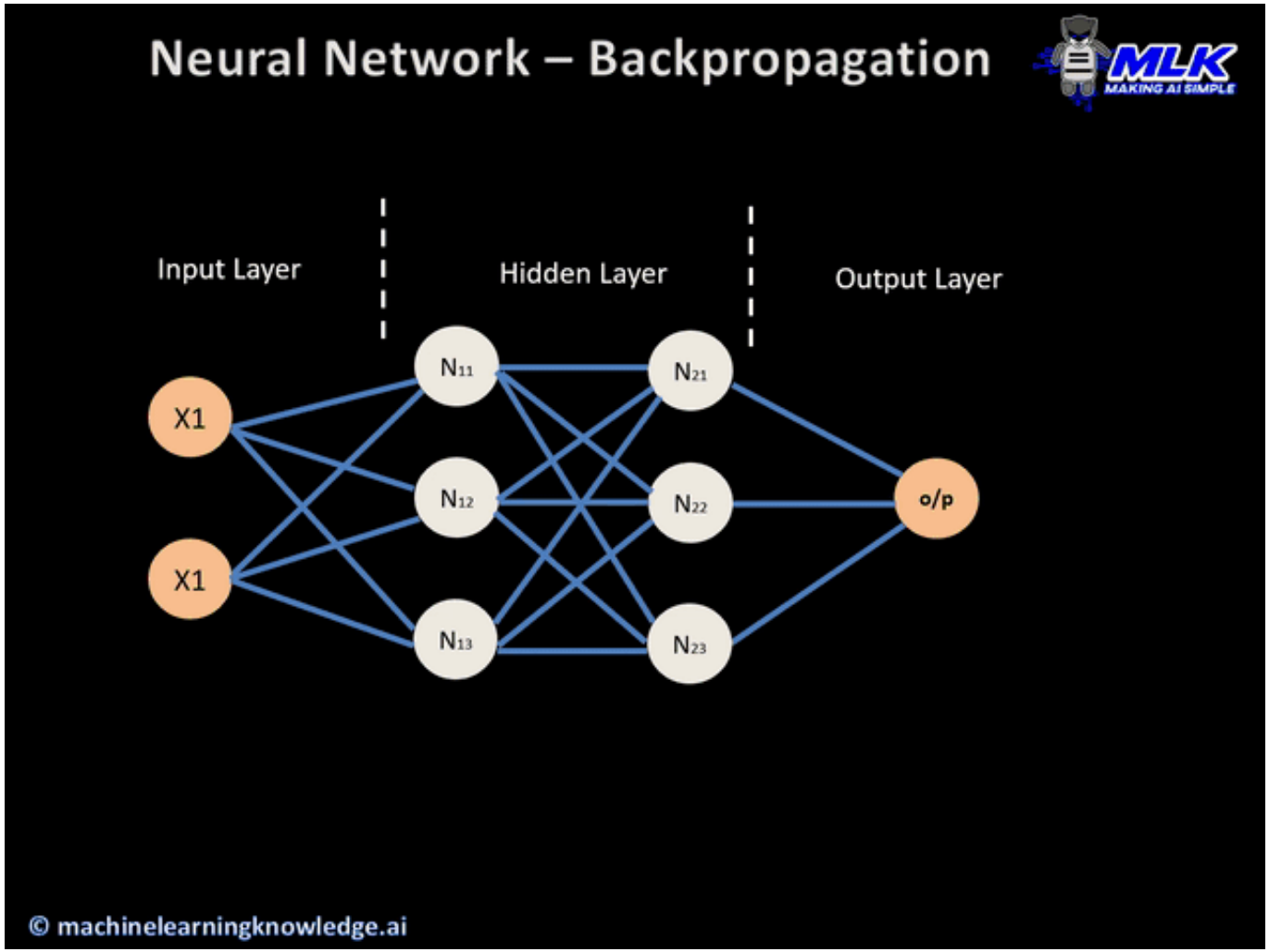
David E. Rumelhart*, Geoffrey E. Hinton† & Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California, San Diego, La Jolla, California 92093, USA
† Department of Computer Science, Carnegie-Mellon University, Pittsburgh, Philadelphia 15213, USA

- David Rumelhart, Geoffrey Hinton, and Ronald Williams publish "Learning representations by back-propagating errors", introducing the method that allows neural networks to learn complex patterns by adjusting internal weights.
- This breakthrough made artificial neural networks viable for practical applications and opened the door for the deep learning revolution of the 2000s and 2010s.

https://www.ibm.com/think/topics/artificial-intelligence

1986: Backpropagation for training neural networks



https://www.ibm.com/think/topics/artificial-intelligence

1997: Deep Blue

IBM's Deep Blue beats then world chess champion Garry Kasparov, in a chess match (and rematch).

https://www.ibm.com/think/topics/artificial-intelligence
https://www.youtube.com/watch?v=KF6sLCeBj0s&ab_channel=BBCNews

2000s: Big Data Era Begins

Why did AI experience rapid advancement in the 21st century rather than in previous decades?

(Internet, personal devices, hardware and software advances)

By this time, the era of big data and high-performance computing is underway, enabling organizations to manage ever-larger data estates, which will one day be used to train AI models.

https://www.ibm.com/think/topics/artificial-intelligence

2010s: Deep Learning Breakthrough

2011
IBM Watson® beats champions Ken Jennings and Brad Rutter at Jeopardy! Also, around this time, data science begins to emerge as a popular discipline.

Credit: IBM

2016
DeepMind's AlphaGo program, powered by a deep neural network, beats Lee Sodol, the world champion Go player, in a five-game match. The victory is significant given the huge number of possible moves as the game progresses (over 14.5 trillion after just four moves). Later, Google purchased DeepMind for a reported USD 400 million.

https://www.theguardian.com/technology/2016/mar/15/alphago-what-does-google-advanced-software-go-next

https://www.ibm.com/think/topics/artificial-intelligence

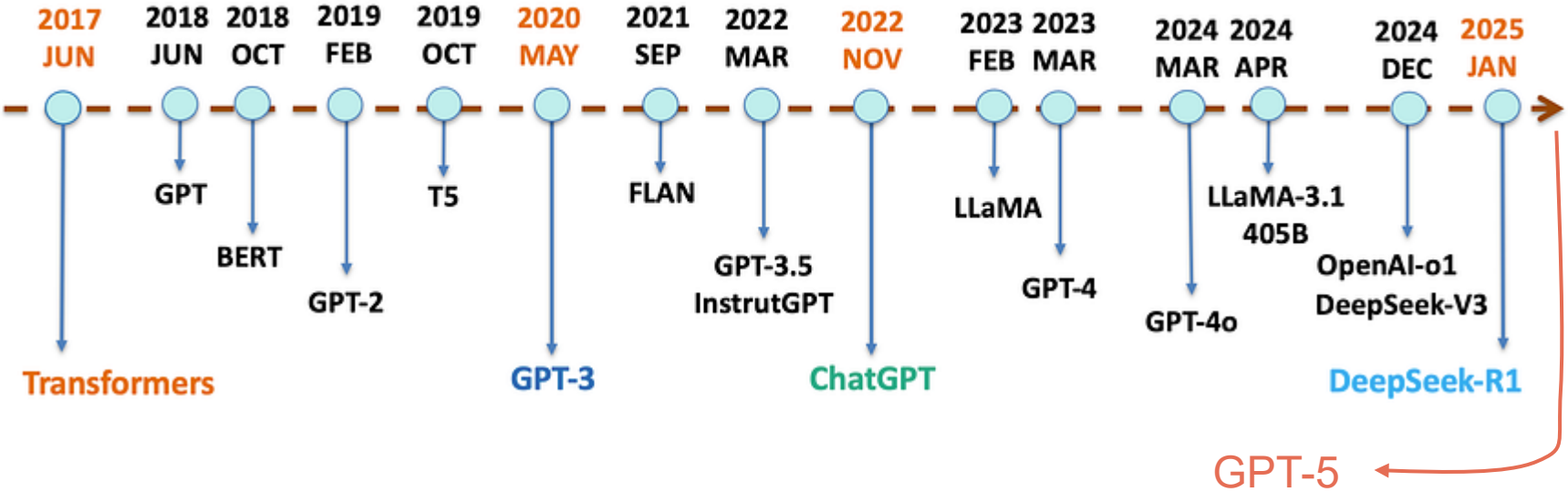
2022 forward : A Rise in Large Language Models

- A rise in large language models or LLMs, such as OpenAI's ChatGPT (where GPT standards for Generative Pre-Trained Transformer), creates an enormous change in performance of AI and its potential to drive enterprise value. With these new generative AI practices, deep-learning models can be pretrained on large amounts of data.

- The latest AI trends point to a continuing AI renaissance. Multimodal models that can take multiple types of data as input are providing richer, more robust experiences. These models bring together computer vision image recognition and natural language processing speech recognition capabilities.

https://www.ibm.com/think/topics/artificial-intelligence

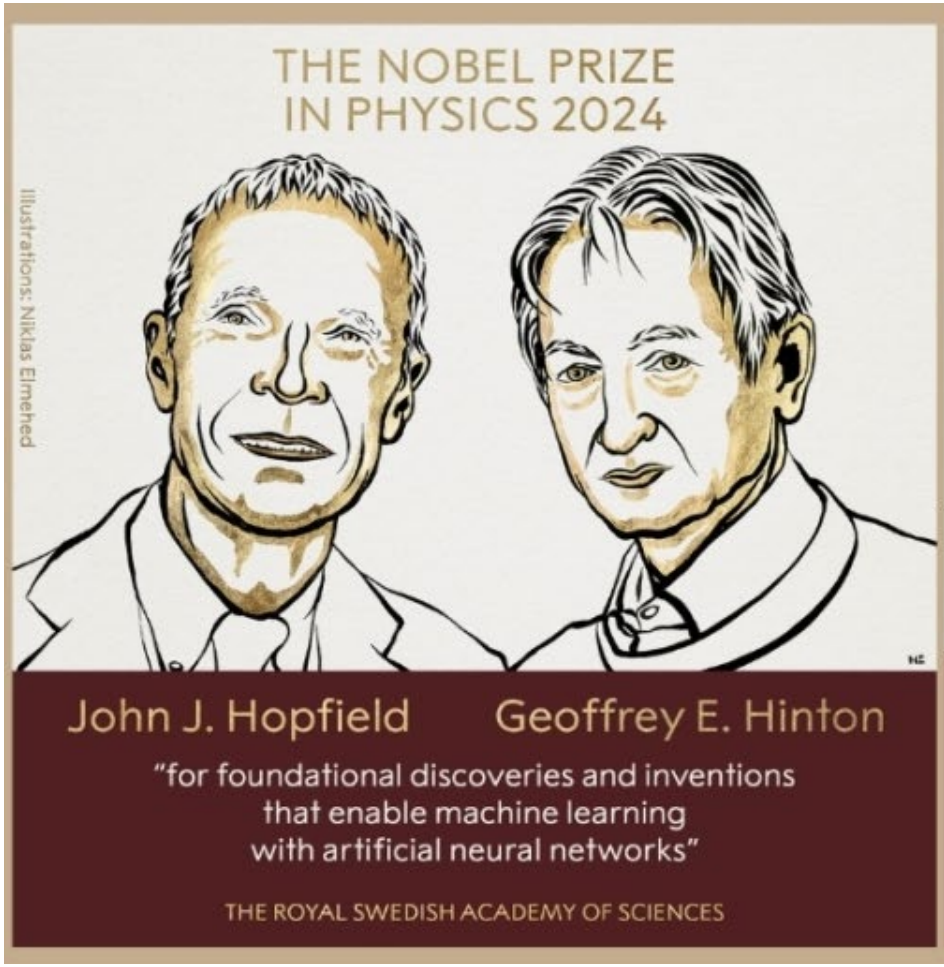
A Brief History of LLMs

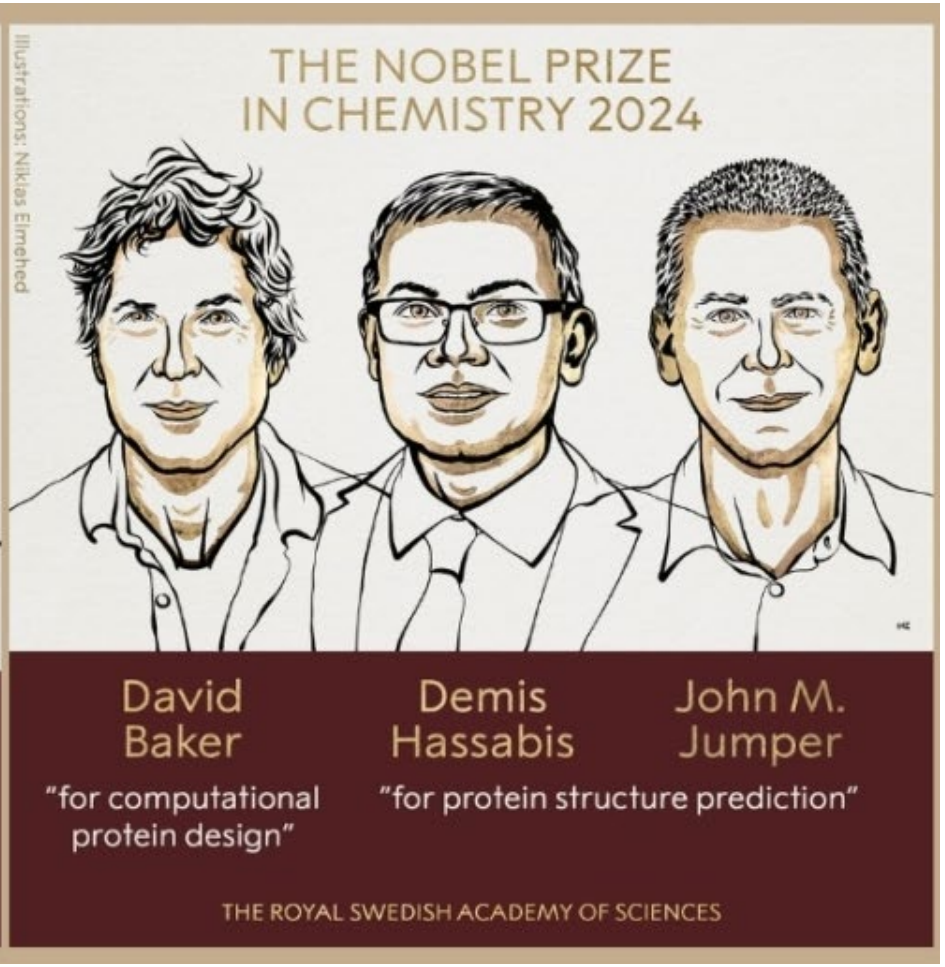


- LLM is arguably the most active area in industry. Many companies have their own “ChatGPT”.
- LLM field is highly competitive, with many service available in addition to ChatGPT (from OpenAI). Examples include Gemini (Google), Perplexity, Claude (Anthropic), Grok (xAI).

https://medium.com/@lmpo/a-brief-history-of-lmms-from-transformers-2017-to-deepseek-r1-2025-dae75dd3f59a

AI Pioneers Won 2024 Nobel Prizes





More about Geoffrey Hinton

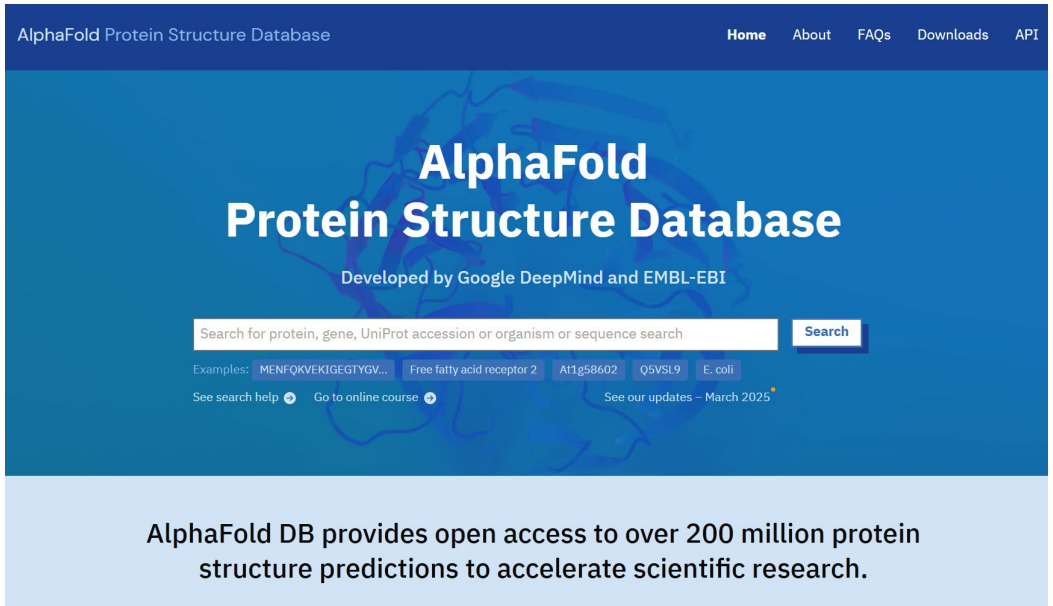
- Known as the "Godfather of AI" for foundational work on artificial neural networks and deep learning.
- Co-authored the seminal 1986 paper popularizing the backpropagation algorithm, essential for training neural networks.
- Co-inventor of Boltzmann machines, and contributed to innovations like AlexNet which revolutionized computer vision in 2012.
- Awarded the 2018 Turing Award with Yann LeCun and Yoshua Bengio for deep learning contributions.
- Worked at University of Toronto and Google Brain; left Google in 2023 to raise awareness about AI risks.

Hinton speaking at the Nobel Prize Lectures in Stockholm in 2024

https://en.wikipedia.org/wiki/Geoffrey_Hinton

AlphaFold: an AI System that Predicts Protein 3D Structures from Amino Acid Sequences

- **AlphaFold 1 (2018)**: Introduced AI-based protein structure prediction; improved speed and accuracy.
- **AlphaFold 2 (2020)**: Major breakthrough with deep learning integration; achieved near-experimental accuracy; outperformed all competitors.
- **AlphaFold 3 (2024):** Expanding capabilities to protein-ligand interactions, RNA folding, and broader applications.



Try it out:

AlphaFold Server

Hemoglobin subunit beta (HBB):

MVHLTPEEKSAVTALWGKVNVDEV
GGEALGRLLVVYPWTQRFFESFGD
LSTPDAVMGNPKVKAHGKKVLGAF
SDGLAHLDNLKGTFAALSELHAHKL
RVDPVNFKLLSHCLLVTLAAHLPAE
FTPAVHASLDKFLASVSTVLTSKYR

https://alphafold.ebi.ac.uk/

Open Discussions

- Where do you encounter AI in your daily life?
- What excites or concerns you most about AI/ML technology in daily life or applications to chemical engineering?